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Completed	Wednesday, 28 October 2026, 9:19 PM
Duration	1 min 56 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It is slower than linear search for small lists
c. It can only be applied to large lists
d. It requires the list to be sorted ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **3**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Linear
c. Both (1) & (3)
d. Hash

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search ends successfully ✓
- b. The search continues in the right sublist
- c. The list is sorted
- d. The search continues in the left sublist

Question **5**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Binary search
- b. Jump search
- c. Hashing
- d. Linear search ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search is more efficient than binary search on large lists.
- c. Linear search requires the list to be sorted.
- d. Linear search divides the list into halves.

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the list is sorted
- c. When the list contains duplicate elements
- d. When the search algorithm finishes

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **9**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By sorting the list first
c. By dividing the list into halves
d. By comparing each element in the list sequentially ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element

Question **12**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100

b. 94 and 99 ✗

c. 89 and 94

d. 90 and 99

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Complexity

c. Bubble ✓

d. Insertion

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓

b. O(n log n)

c. O(log n)

d. O(1)

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the left sublist ✓
c. In the right sublist
d. At the beginning of the list